

Project Workflow: Week 5

Phase: Ball Movement & Paddle Implementation (Bounce Ball Game)

1. Objective

The primary objective of Week 5 was to implement continuous ball movement mechanics and player-controlled paddle responsiveness for the Bounce Ball Game. Building upon the Week 4 layout, this phase focused on establishing autonomous ball movement, enabling keyboard inputs for paddle controls, and configuring the timer-driven game loop.

2. Detailed Task Breakdown

- **Ball Movement Mechanics:** Enable continuous ball position updating across the game canvas.
- **Paddle Implementation:** Created the player paddle component positioned at the bottom of the screen with customizable width, height, and rendering parameters.
- **Keyboard Input Listener:** Implemented `KeyListener` events (Left/Right Arrow keys and A/D keys) to capture real-time user input for smooth horizontal paddle movement.
- **Boundary Restraints:** Applied positional constraints on the paddle to ensure it remains strictly within the screen bounds during movement.
- **Animation & Game Loop Setup:** Integrated `javax.swing.Timer` to maintain a smooth frame rate (~60 FPS) for continuous rendering of ball trajectory and paddle movement.

3. Deliverables

- **Autonomous Ball Movement:** Continuous ball displacement mechanism driven by velocity vectors and frame repaints.

- **Movable Player Paddle:** Horizontal paddle movement powered by smooth keyboard controls and canvas edge clamping.
- **Key Input Controller:** Event-driven key listener integration supporting multiple movement key bindings.
- **Game Loop Engine:** Timer-controlled animation system ensuring consistent rendering and state updates.

4. Tools & Resources

- **Development Environment:** Java JDK 11+, VS Code / IntelliJ IDEA / Eclipse
- **Libraries & Technologies:** Java Swing (`javax.swing.Timer`, `JPanel`, `JFrame`), Java AWT Event Handling (`KeyListener`, `KeyEvent`), Graphics2D Rendering Engine